

PUGAZHMANI K

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[LinkedIn](#) — [GitHub](#)

PROFESSIONAL SUMMARY

Final-year B.E. Computer Science and Engineering (Cybersecurity) student with hands-on experience developing machine learning, LLM, RAG, AI automation, backend, and security-focused systems. Skilled in Python, AI/ML, FastAPI, REST APIs, vector databases, local LLM deployment, data processing, Docker, PostgreSQL, and Git. Experienced in designing practical AI applications, integrating intelligent systems into software platforms, and developing security-focused AI solutions.

TECHNICAL SKILLS

Programming: Python, Go, C, SQL

AI / Machine Learning: PyTorch, Llama 3, RAG, NLP, XGBoost, Diffusion Models, Vector Databases, ChromaDB

AI Engineering: LLM Integration, Local LLM Deployment, Prompt Engineering, Semantic Search, Document Processing, Data Extraction, AI Application Development, Explainable AI

Backend / APIs: FastAPI, REST APIs, API Integration, Asynchronous Backend Development

Data / Databases: PostgreSQL, SQLite, ChromaDB, Embeddings, Data Processing

DevOps / Development: Docker, Docker Compose, Git, GitHub

Frontend: React.js, Vite, Tailwind CSS, Streamlit

Cybersecurity: WAF, SIEM, Threat Telemetry, Vulnerability Analysis, Network Forensics, Malware Analysis

PROFESSIONAL EXPERIENCE

CyberWolf — AI & Software Engineer

2024–2026

- Developed AI-driven software systems focused on LLM applications, Retrieval-Augmented Generation (RAG), automation, intelligent security systems, and AI-enabled application development.
- Architected and implemented practical AI systems using Python, FastAPI, local LLMs, vector databases, REST APIs, and containerized infrastructure.
- Designed reusable software components and backend services for AI and cybersecurity applications, integrating data processing and intelligent automation workflows.
- Researched and experimented with LLM architectures, RAG pipelines, machine learning models, AI integration techniques, and AI security applications.
- Collaborated on technical workshops, hackathons, engineering initiatives, and knowledge-sharing activities related to AI and cybersecurity.

SELECTED AI & ENGINEERING PROJECTS

CyberWolf VulnStream

Python, FastAPI, RAG, ChromaDB, LLMs

- Developed a cybersecurity-focused Retrieval-Augmented Generation (RAG) platform that centralizes threat intelligence from multiple trusted sources into a unified knowledge system.
- Designed data ingestion and normalization workflows to transform heterogeneous cybersecurity intelligence into a consistent retrieval-ready format.
- Implemented semantic search and vector-based retrieval using embeddings and ChromaDB to provide contextual threat intelligence for LLM-powered workflows.
- Integrated cybersecurity knowledge sources including CVE, CWE, CAPEC, and MITRE ATT&CK data for AI-assisted security research.

X-MalForensics

Python, XGBoost, SHAP, Explainable AI, Hugging Face

- Developed an Explainable AI framework for Windows malware detection using an XGBoost classifier trained on the EMBER 2018 dataset.
- Integrated SHAP-based explainability to generate human-readable feature-level explanations for malware classification decisions.
- Designed a forensic analysis workflow to help security analysts understand model predictions rather than relying solely on black-box classification.
- Published the trained model on [Hugging Face](#) for model access and inference.

CuraCore

Python, FastAPI, React, Local LLMs, SQLite

- Developed an offline AI-enabled hospital management system using locally hosted Large Language Models to keep sensitive data processing on local infrastructure.
- Integrated open-weight AI models for clinical summarization and structured data extraction without relying on third-party cloud AI APIs.
- Built a responsive React frontend connected to an asynchronous FastAPI backend for end-to-end application workflows.

Healthy Buddy+

Python, Streamlit, Llama 3, RAG

- Engineered a Retrieval-Augmented Generation (RAG) application using Llama 3 to provide context-aware answers from proprietary PDF documents.
- Developed document ingestion, chunking, embedding, and semantic similarity search workflows for knowledge retrieval.
- Implemented citation-backed responses to improve traceability and grounding of retrieved information.

WolfGuard 360

Golang, Python, FastAPI, React, Docker, PostgreSQL

- Architected a containerized Security-as-a-Service platform with a reverse-proxy Web Application Firewall (WAF) for intercepting malicious HTTP traffic.
- Engineered an asynchronous threat telemetry pipeline using FastAPI to route security events into a centralized monitoring dashboard.
- Implemented multi-tenant JWT authentication and API key revocation to provide tenant isolation and controlled access.

EDUCATION

B.E. Computer Science and Engineering (Cybersecurity) — **SKP Engineering College** Expected Graduation: 2027
Affiliated to Anna University CGPA: 8.0

ACHIEVEMENTS & LEADERSHIP

- Secured top-three placements in 10+ national-level hackathons, including Hack India, IFET Hackathon, and Taksha Sheela University Hackathon.
- Participated in and presented at 50+ national-level technical conferences and engineering events.
- Project Lead — “AgroMed”, IFET Hack Warriors, Jan 2026; contributed to solution planning and architecture for agriculture and healthcare analytics.
- Core Member — “AANIVARU AKKUVARU”, CyberWolf / Anniveru Akkuveru Team, Oct 2025; contributed to system architecture and digital learning initiatives.

CERTIFICATIONS

- Certified Online Fraud Prevention Specialist (COFPS) — Hackfix Academy

AREAS OF INTEREST

Artificial Intelligence, Machine Learning, LLM Applications, Retrieval-Augmented Generation, AI Automation, Backend Engineering, Explainable AI, AI Security, Threat Intelligence